

GENIE3 candidate multiomics evidence

Rank	Candidate regulator	TCR activation mRNA	Incremental BTLA mRNA	Protein abundance	Phosphosite abundance	Independent association
1	EGR2	↑ 1h, 4h, 24h	No qualifying change	—	—	—
2	JUNB ●	↑ 1h, 24h	No qualifying change	—	—	—
3	FOXP1	No qualifying change	No qualifying change	No qualifying change	—	No qualifying change
4	HIVEP3	↑ 4h, 24h	No qualifying change	—	—	—
5	IRF8 ●	No qualifying change	No qualifying change	—	—	—
6	STAT1 ●	No qualifying change	No qualifying change	No qualifying change	—	No qualifying change
7	BHLHE40	No qualifying change	No qualifying change	—	—	—
8	PLAGL2 ●	No qualifying change	No qualifying change	No qualifying change	—	No qualifying change
9	ZBTB32 ●	No qualifying change	No qualifying change	—	—	—
10	CYCS	No qualifying change	No qualifying change	No qualifying change	—	No qualifying change
11	MDM2	No qualifying change	No qualifying change	—	—	—
12	GAR1	No qualifying change	No qualifying change	No qualifying change	—	No qualifying change
13	MYC	No qualifying change	No qualifying change	No qualifying change	4h, 24h (3 sites)	Phosphosite
14	ZEB2	No qualifying change	No qualifying change	—	—	—
15	FOXM1	No qualifying change	No qualifying change	—	—	—
16	SMAP2	No qualifying change	No qualifying change	4h	—	Protein
17	NR4A2	No qualifying change	No qualifying change	—	—	—
18	CREM ●	No qualifying change	No qualifying change	—	—	—
19	ID2	↑ 4h	No qualifying change	—	—	—
20	STAT5B	No qualifying change	No qualifying change	20min	—	Protein
21	TRAF4	No qualifying change	No qualifying change	—	—	—
22	GTF2A2	No qualifying change	No qualifying change	—	—	—
23	TGIF1	No qualifying change	No qualifying change	—	—	—
24	MAF	Failed QC	No qualifying change	—	—	—
25	PRDM1 ●	↑ 4h	No qualifying change	—	—	—

Contextual mRNA (qualifying)

Independent protein / phosphosite

No qualifying change

— absent / not assessable

● shared with the other model's top 25

Failed QC = quality-control failure

Candidate mRNA is contextual (TCR vs unstimulated control; BTLA+TCR vs TCR) and is shown in blue. Protein and phosphosite are independent observed molecular layers (BTLA+TCR vs TCR; source thresholds) and are shown in green. The combined BTLA+TCR vs unstimulated-control mRNA contrast was not available from the fitted source models and was not reconstructed. Direction is not interpreted as supportive or opposing (unsigned rankings). Row order matches Table 4B exactly

Figure | GENIE3 top 25 candidate regulator multiomics evidence summary.